SE-Week-15 (1).pptx [slide 1 - 12]

1. Quality Management

Quality Management is the process of ensuring that software:

- **Meets required standards**
- **Meets customer needs**
- **Follows organizational goals**
- **Has acceptable quality**

It acts as an **independent check** on software development.

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2. Quality Management Activities

- Checks project deliverables
- Ensures standards are followed
- Ensures software quality is maintained
- Verifies consistency with organizational goals

Important Point

The **quality team should be independent** from the development team.

- Gives objective evaluation
- Avoids bias
- Reports quality honestly

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3. Software Quality Attributes

Quality is not only about correctness.

- Reliability → Software **works correctly** over time.
- Robustness → Software **handles invalid inputs and error**s.
- Resilience → Software **recovers from failures.**
- Safety → Software **does not cause harm**.
- Security → Protects against **unauthorized access.**
- Understandability → Easy to understand.
- Testability → Easy to test.
- Adaptability → **Easy to modify** for new requirements.
- Modularity → Software divided into **independent modules**.
- Complexity → Lower complexity = Better quality.
- Portability → Software works on **different platforms.**
- Usability → Easy for users to operate.
- Reusability → Components can be reused in other projects.
- Efficiency → Uses minimum resources
- Learnability → Easy for new users to learn.

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4. Quality Conflicts

- A system **cannot be optimized for every quality attribute at the same time.**
- Improving one quality may reduce another.
- **Example**
  - Increasing robustness:
  - ✔ Better error handling
  - ❌ May reduce performance

Therefore:

- **Prioritize important qualities**

- Make trade-offs

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5. Quality Plan

A Quality Plan specifies:

- **Most important quality attributes**
- Quality goals
- Quality assessment process

It tells: **How quality will be measured and checked.**

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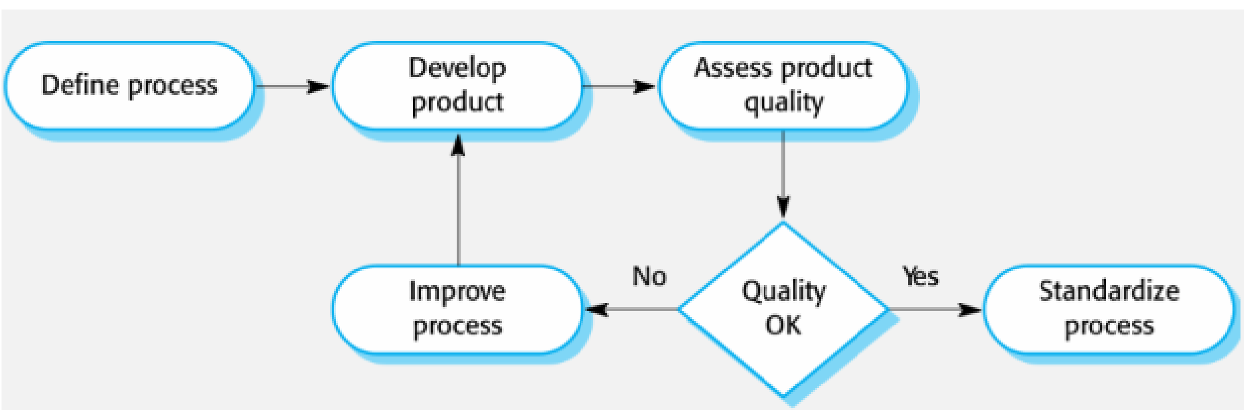
6. Process-Based Quality



Main Idea: **Good process → Good software quality.**

If development follows:

- Standards
- Reviews
- Testing procedures

then software quality usually improves.

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7. Software Standards

Standards are **rules** that define:

- **Required product attributes**
- Required **development processes**

Standards **help maintain quality.**

› Types of Standards

International Standards

- Used worldwide.
- Example: ISO 9001.

National Standards

- Used within a country.

Organizational Standards

- Created by a company.

Project Standards

- Created specifically for one project.

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8. Importance of Standards

1. Best Practices

- Standards **contain proven methods**.
- **Benefits**: **Avoid repeating old mistakes**

2. Define Quality

- Help organization decide: What does quality mean?

3. Continuity

- New employees can understand work procedures quickly by understanding the standards that are used

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9. Product Standards vs Process Standards

Product Standards: Apply to software products

include document standards, eg; structure of requirements documents, documentation standards, such as a standard comment header for an object class definition, and coding standards, which define how a programming language should be used.

- Requirements document structure
- Method header format
- Java coding style
- Project plan format

- Change request form

Focus: "What should the product look like?"

Process Standards: Apply to development activities

Process that should be followed during development

- Design review process
- Version release process
- Project approval process
- Change control process
- Test recording process

Focus: "How should the work be done?"

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10. Problems with Standards

Developers may think standards are:

- Old
- Irrelevant
- Too restrictive

As technology changes, **standards must be updated.**

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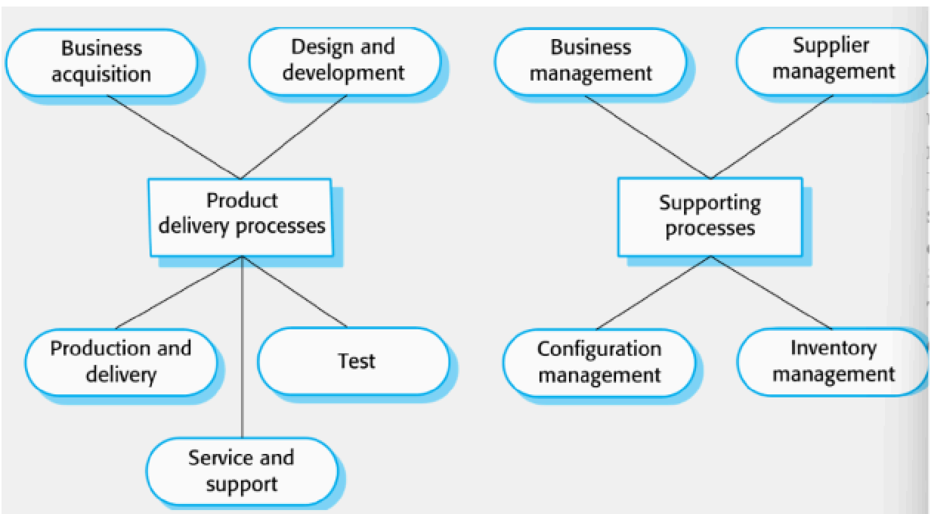
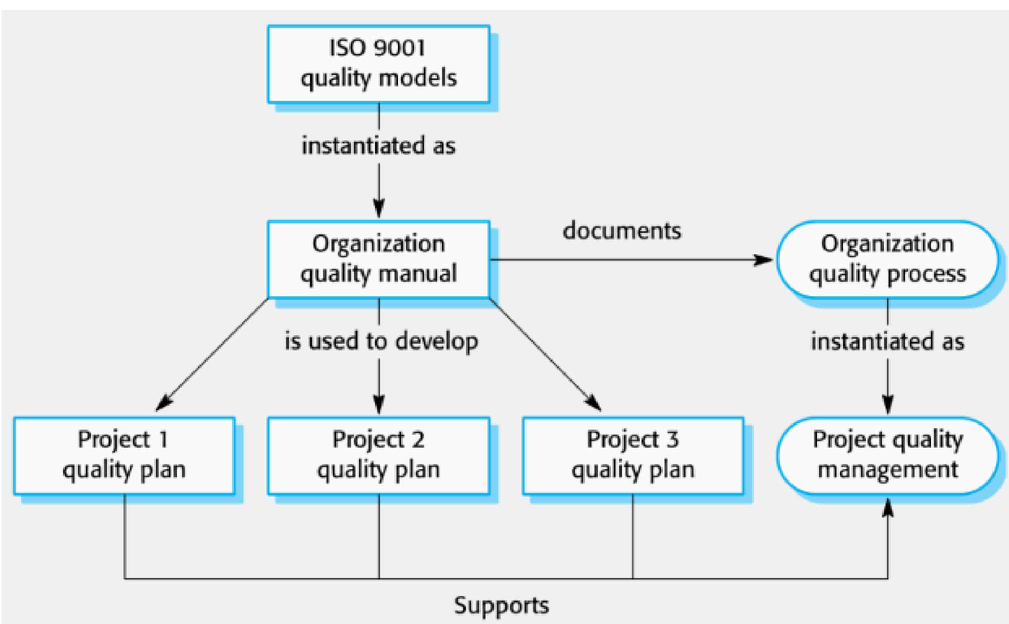
11. ISO 9001

- ISO 9001 is an **international quality standard**

- Used as a framework for creating quality management systems.
- It applies to organizations that:
  - Design products
  - Develop products
  - Maintain products
  - Including software organizations.





Purpose of ISO 9001

Helps organizations:

- Maintain quality
- Standardize processes
- Improve consistency
- Improve customer confidence

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12. ISO 9001 Certification

To become certified:

Step 1: Create a Quality Manual.

The manual documents:

- Quality procedures
- Standards
- Processes

Step 2: External auditors inspect organization.

Step 3: If requirements are met: Organization receives ISO certification.

Why Certification Matters?

Many customers prefer or require certified suppliers because it shows commitment to quality.
